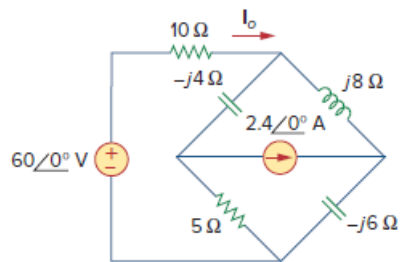


Problem

Calculate current I_o in the circuit of Fig. 10.11.

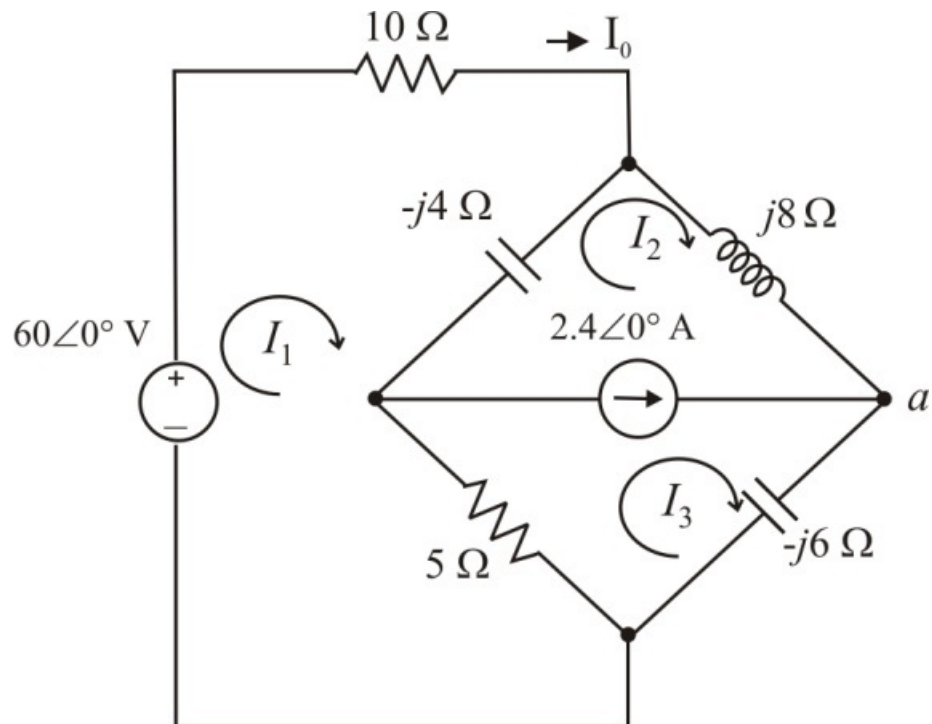
Figure 10.11:



Step-by-step solution

Step 1 of 5

Consider the following circuit diagram:



Step 2 of 5

Figure 1

Step 3 of 5

Assume that currents $\mathbf{I}_1$, $\mathbf{I}_2$ and $\mathbf{I}_3$ are flowing in the circuit loops.

The following is the current at point a from the circuit:

$$\mathbf{I}_3 = \mathbf{I}_2 + 2.4\angle 0^\circ$$

Apply Kirchhoff's voltage law to mesh 1.

$$10\mathbf{I}_1 - j4(\mathbf{I}_1 - \mathbf{I}_2) + 5(\mathbf{I}_1 - \mathbf{I}_3) = 60$$

$$\mathbf{I}_1(15 - j4) + j4\mathbf{I}_2 - 5\mathbf{I}_3 = 60$$

$$\mathbf{I}_1(15 - j4) + j4\mathbf{I}_2 - 5(\mathbf{I}_2 + 2.4\angle 0^\circ) = 60$$

$$\mathbf{I}_1(15 - j4) + \mathbf{I}_2(-5 + j4) = 72 \dots\dots (1)$$

A super mesh forms between mesh 2 and mesh 3 due to the presence of current source.

Write the Kirchhoff's voltage law equation for super mesh:

$$j8\mathbf{I}_2 - j6\mathbf{I}_3 + 5(\mathbf{I}_3 - \mathbf{I}_1) - j4(\mathbf{I}_2 - \mathbf{I}_1) = 0$$

$$(-5 + j4)\mathbf{I}_1 + j4\mathbf{I}_2 + (5 - j6)\mathbf{I}_3 = 0$$

$$(-5 + j4)\mathbf{I}_1 + j4\mathbf{I}_2 + (5 - j6)(\mathbf{I}_2 + 2.4\angle 0^\circ) = 0$$

$$(-5 + j4)\mathbf{I}_1 + (5 - j2)\mathbf{I}_2 = -12 + j14.4 \dots\dots (2)$$

Step 4 of 5

Convert equations (1) and (2) into matrix form as follows.

$$\begin{bmatrix} 15 - j4 & -5 + j4 \\ -5 + j4 & 5 - j2 \end{bmatrix} \begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix} = \begin{bmatrix} 72 \\ -12 + j14.4 \end{bmatrix}$$

Calculate the determinant values of matrices to find $\mathbf{I}_1$ and $\mathbf{I}_2$.

$$\begin{aligned} \Delta &= \begin{vmatrix} 15 - j4 & -5 + j4 \\ -5 + j4 & 5 - j2 \end{vmatrix} \\ &= (15 - j4)(5 - j2) - (-5 + j4)(-5 + j4) \\ &= (75 - j30 - j20 - 8) - (25 - j20 - j20 - 16) \\ &= 58 - j10 \end{aligned}$$

$$\begin{aligned} \Delta_1 &= \begin{vmatrix} 72 & -5 + j4 \\ -12 + j14.4 & 5 - j2 \end{vmatrix} \\ &= 72(5 - j2) - (-5 + j4)(-12 + j14.4) \\ &= 360 - j144 - (60 - j72 - j48 - 57.6) \\ &= 357.6 - j24 \end{aligned}$$

Step 5 of 5

Calculate the current $\mathbf{I}_o$ in mesh 1:

$$\begin{aligned} \mathbf{I}_o &= \mathbf{I}_1 \\ &= \frac{\Delta_1}{\Delta} \\ &= \frac{357.6 - j24}{58 - j10} \\ &= 6.089\angle 5.94^\circ \text{ A} \end{aligned}$$

Therefore, the value of current $\mathbf{I}_o$ is $\boxed{6.089\angle 5.94^\circ \text{ A}}$.